

Question ID: 294db8ec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which of the following is equivalent to $2x^3 + 4$?

Answer

- A. $4(x^3 + 4)$
- B. $4(x^3 + 2)$
- C. $2(x^3 + 4)$
- D. $2(x^3 + 2)$

Correct Answer: D

Rationale

Choice D is correct. The expression $2x^3 + 4$ has two terms, $2x^3$ and 4. The greatest common factor of these two terms is 2. Factoring 2 from each of these terms yields $2(x^3) + 2(2)$, or $2(x^3 + 2)$.

Choices A and B are incorrect because 4 is not a factor of the term $2x^3$. Choice C is incorrect and may result from factoring 2 from $2x^3$ but not from 4.